

Question 1

Consider a resistive-load inverter circuit with $V_{DD}=5\text{ V}$, $k'_n=20\text{ }\mu\text{A/V}^2$, $V_{T0}=0.8\text{ V}$, $R_L=200\text{ k}\Omega$, and $W/L=2$.

Calculate the critical voltages ($V_{OL}, V_{OH}, V_{IL}, V_{IH}$) on the VTC and find the noise margins of the circuit.

Question 2

Design a resistive-load inverter with $R=1\text{ k}\Omega$, such that $V_{OL}=0.6\text{ V}$. The enhancement-type nMOS driver transistor has the following parameters:

$$V_{DD}=5\text{ V},$$

$$V_{T0}=1\text{ V},$$

$$\gamma=0.2\text{ V}^{1/2},$$

$$\lambda=0,$$

$$\mu_n C_{ox}=22.0\text{ }\mu\text{A/V}^2$$

(a) Determine the required aspect ratio, W/L . $\leftarrow 90$

(b) Determine V_{IL} and V_{IH} . $V_{IL}=1.505\text{ V}, V_{IH}=3.09\text{ V}$

(c) Determine noise margins NM_L and NM_H . $\leftarrow 1.91\text{ V}$

Question 3

Consider a CMOS inverter circuit with the following parameters:

$$V_{DD}=3.3\text{ V},$$

$$V_{T0,n}=0.6\text{ V},$$

$$V_{T0,p}=-0.7\text{ V},$$

$$k_n=200\text{ }\mu\text{A/V}^2,$$

$$k_p=80\text{ }\mu\text{A/V}^2$$

$$\begin{aligned} V_{IL} &= 1.07\text{ V} \\ V_{IH} &= 1.55\text{ V} \\ NM_L &= 1.07\text{ V} \\ NM_H &= 1.75\text{ V} \end{aligned}$$

Calculate the noise margins of the circuit. Notice that the CMOS inverter being considered here has $k_R = 2.5$ and $V_{T0,n} \neq |V_{T0,p}|$; hence, it is not a symmetric inverter.

Question 4

Consider a CMOS inverter with the following parameters:

$$\text{nMOS} \quad V_{T0,n}=0.6\text{ V}, \quad \mu_n C_{ox}=60\text{ }\mu\text{A/V}^2, \quad (W/L)_n=8$$

$$\text{pMOS} \quad V_{T0,p}=-0.7\text{ V}, \quad \mu_p C_{ox}=25\text{ }\mu\text{A/V}^2, \quad (W/L)_p=12$$

Calculate the noise margins and the switching threshold (V_{th}) of this circuit. The power supply voltage is

$$V_{DD}=3.3\text{ V}.$$

$$\begin{aligned} V_{IL} &= 1.16\text{ V} \\ V_{IH} &= 1.69\text{ V} \\ NM_L &= 1.16\text{ V} \\ NM_H &= 1.61\text{ V} \\ V_{th} &= 1.48\text{ V} \end{aligned}$$

Question 5

Design of a CMOS inverter circuit:

Use the parameters given below:

$$\text{nMOS} \quad V_{T0,n}=0.6 \text{ V}, \quad \mu_n C_{ox}=60 \text{ } \mu\text{A/V}^2, \quad (W/L)_n=8$$

$$\text{pMOS} \quad V_{T0,p}=-0.7 \text{ V}, \quad \mu_p C_{ox}=25 \text{ } \mu\text{A/V}^2, \quad (W/L)_p=12$$

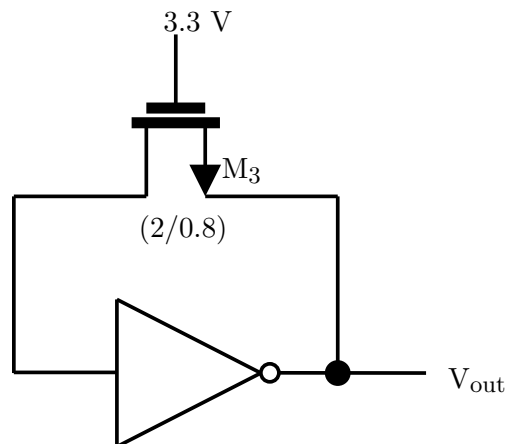
The power supply voltage is $V_{DD}=3.3 \text{ V}$. The channel length of both transistors is $L_n=L_p=0.8 \text{ } \mu\text{m}$.

(a) Determine the (W_n/W_p) ratio so that the switching (inversion) threshold voltage of the circuit is $V_{th}=1.4 \text{ V}$.
 $\uparrow 0.9375$

(b) The CMOS fabrication process used to manufacture this inverter allows a variation of the $V_{T0,n}$ value by $\pm 15\%$ around its nominal value, and a variation of the $V_{T0,p}$ value by $\pm 20\%$ around its nominal value. Assuming that all other parameters (such as μ_n , μ_p , C_{ox} , W_n , W_p) always retain their nominal values, find the upper and lower limits of the switching threshold voltage (V_{th}) of this circuit.

Question 6
 $\uparrow 1.51 \text{ V} \quad \uparrow 1.29 \text{ V}$

Consider the CMOS inverter designed in **Question 5**, with the following circuit configuration:



- (a) Calculate the output voltage level V_{out} . $V_{out} = V_{in} = V_{th}$
- (b) Determine if the process-related variation of $V_{T0,n}$ of M_3 has any influence upon the output voltage V_{out} . No effect.
- (c) Calculate the total current being drawn from the power supply source, and determine its variation due to process-related threshold-voltage variations.

$$I = \frac{K_n}{2} (V_{th} - V_{T0,n})^2$$

$$= 19.2 (W/L)_n \text{ } \mu\text{A}$$